Exp 8

Server

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <time.h>
#define PORT 8080
#define BUFFER_SIZE 1024
int main()
{
    int sockfd;
    char buffer[BUFFER_SIZE];
    struct sockaddr_in server_addr, client_addr;
    socklen_t len;
    sockfd = socket(AF_INET, SOCK_DGRAM, 0);
    server_addr.sin_family = AF_INET;
    server_addr.sin_addr.s_addr = INADDR_ANY;
    server_addr.sin_port = htons(PORT);
    bind(sockfd, (struct sockaddr *)&server_addr, sizeof(server_addr));
    printf("UDP Time Server Running...\n");
    while (1)
    {
        len = sizeof(client_addr);
        recvfrom(sockfd, buffer, BUFFER_SIZE, 0,
                 (struct sockaddr *)&client_addr, &len);
        time_t current_time;
        time(&current_time);
        strcpy(buffer, ctime(&current_time));
        sendto(sockfd, buffer, strlen(buffer), 0,
               (struct sockaddr *)&client_addr, len);
    }
    close(sockfd);
    return 0;
}
```

Client

```
#include <stdio.h>

#include <stdlib.h>

#include <string.h>

#include <unistd.h>

#include <arpa/inet.h>

#define PORT 8080

#define BUFFER_SIZE 1024

int main()

{

    int sockfd;

    char buffer[BUFFER_SIZE] = "TIME";

    struct sockaddr_in server_addr;

    socklen_t len;

    sockfd = socket(AF_INET, SOCK_DGRAM, 0);

    server_addr.sin_family = AF_INET;

    server_addr.sin_port = htons(PORT);

    server_addr.sin_addr.s_addr = inet_addr("127.0.0.1");

    sendto(sockfd, buffer, strlen(buffer), 0, (struct sockaddr *)&server_addr, sizeof(server_addr));

    len = sizeof(server_addr);

    recvfrom(sockfd, buffer, BUFFER_SIZE, 0, (struct sockaddr *)&server_addr, &len);

    printf("Server Time: %s\n", buffer);

    close(sockfd);

    return 0;

}
```

Server

```
anandhu@anan:/mnt/e/kichu/Academic/CN lab/exp8$ ./serv
UDP Time Server Running...
```

Client

```
anandhu@anan:/mnt/e/kichu/Academic/CN lab/exp8$ ./cli
Server Time: Thu Oct 22 13:19:30 2026
```